

3 An Evolutionary Approach to Emotions in Sports

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Emotions play an important role in sports, as in all areas of life. The desire to experience or not to experience an emotion motivates a lot of behaviour in humans (Ekman, 2007). While most people would claim that they have an intuitive understanding about emotions and use the term frequently in their everyday language, science has struggled in dealing with emotions and there is currently little consensus on how to define and measure emotions (Beck, 2015). Therefore, evidence-based guidelines for sports are still hard to derive from research in the area of emotions. This is further complicated by the observation from Richard Lazarus on emotion research in sport:

In the past, two almost separate research literatures have grown up, one centered on stress and coping, the other on emotion. [...] This is illogical and counterproductive. Stress is important in its own right, but emotion encompasses all of the important phenomena of stress. I believe the emotions provide a far richer understanding of the adaptational struggles of human and infrahuman animals.

(Lazarus, 2000, p. 231)

Unfortunately, the majority of research on emotions in sports has narrowly focused on the emotions of anxiety or stress and has led to potential theoretical misunderstandings on the role of emotions in influencing peoples' behaviour in sports (Lazarus, 2000). Hence, this chapter aims to provide an up-to-date synthesis of cross-disciplinary emotion theory and research in setting up a framework to study emotions in sport and derive evidence-based recommendations in applied sport settings.

Emotion Theory

Nothing in biology makes sense except in the light of evolution.

(Wilson, 2019, p. 5)

It has recently been suggested that an evolutionary approach to emotions might help to address the lack of consensus in this field of study (Al Shawaf, Conroy-Beam, Asao, & Buss, 2016) as this perspective allows one to think

about emotions from a broader perspective. Charles Darwin (1872) inspired this big picture approach to emotions that has proven useful to integrate piecemeal research findings. Darwin's general take on emotions was to regard emotions as bodily adaptations (e.g. specialised brain networks) that transform sensory information into adaptive behaviour. Darwin's definition of evolution as descent with modification implies that evolution rarely creates something completely new, but instead old traits are refurbished and modified to suit current needs in a given species. Regarding emotions, this means that it is unlikely that any of the emotions are uniquely human. Therefore, the difference between humans and animals regarding their emotional lives is one of degree rather than of kind (Darwin, 1871). In fact, the brains of humans and other animals are so similar that

we study fear in the rat's amygdala. Dogs trained to lie still in a brain scanner show activity in the caudate nucleus when they expect a hot dog in the same way this region lights up in businessmen who are promised a monetary bonus.

(de Waal, 2019, p. 50)

Evolutionary inspired theories regard emotions to be specialised neuronal networks that evolved to prepare the organism for different categories of recurring situations/stimulus patterns in the environment and the body. More specifically, Ekman (2007, 2016) described emotions as a complex pattern of physiological and psychological changes that are adapted to meaningful situations. These changes include physiological arousal, feelings, cognitive processes, expressions, behavioural tendencies, and overt behaviour, that equip humans and other animals with "emotional hardware" to respond adaptively to recurring problems and opportunities in their environment (e.g. threat, disgust, or feelings of sexual arousal). Or as de Waal (2019, p. 85) puts it, "emotions help us navigate a complex world that we don't fully comprehend by combining individual experience with assessment of the environment to prepare the organism for the optimal response". In this respect, emotions have the great advantage over instincts as they do not dictate rigid behaviour, instead they focus the mind and prepare the body while leaving room for experience and judgement.

As a comprehensive account of the psychological and physiological changes that coincide with specific emotions is beyond the scope of this chapter, I will only briefly describe the most relevant physiological, cognitive, and expressive features of emotions for the context of sports. Following Ekman (2007), an emotion is assumed to be a very brief episode and is typically experienced as happening to us, instead of chosen by us. Longer lasting emotional experiences are not conceptualised as emotions, but as moods. Important components of emotions are cognitive appraisal processes, in which people automatically scan the environment for things that are of importance to them. People only become aware of an emotion when these

automatic appraisal processes are complete. An important distinction has to be made between the physical reactions of emotions and feelings that arise from thoughts. Feelings arise when emotions “penetrate” consciousness (Beckoff, 2007) or, stated differently, feelings are the conscious cognitive evaluations of the somatic emotional reactions to triggering environmental stimuli. A lot of the debate on emotions in the psychological literature (e.g. Barrett, 2006) seems unnecessary as soon as one draws a sharp line between emotions and feelings (de Waal, 2019). Unfortunately, there is no simple one-on-one mapping between emotions and feelings as every language has its own concepts to describe subjective states, and people bring different experiences to how they feel and why. After people become aware that they are experiencing a feeling, they can, if desired, reappraise the situation and potentially change the nature and intensity of the feeling.

Moreover, it has been suggested that there are certain universal emotional themes (i.e. basic emotions; for example fear, anger, disgust, joy, and sadness; see Table 3.1 for a recent summary of the key tenets of basic emotion theory) that reflect our evolutionary history (Ekman, 1992), in addition to many culturally learned variations that reflect our individual experiences. However, other emotion researchers have argued against distinct categories of emotions and advocate a dimensional approach that typically involves the dimensions of pleasant–unpleasant and low–high intensity or activation (Gendron, Crivelli, & Barrett, 2018; see Cowen & Keltner, 2017 for a recent integration of these competing theories).

Classifying emotional experiences has been a huge problem in emotion science, which is nicely captured in the following quote: “The never-ending debate about how to classify emotions, or even what emotion is, reminds me of a phase in biology when our main preoccupation was the classification of plants and animals” (de Waal, 2019, p. 169). This problem is also evident in the sports literature, which is why the only textbook on emotions in sports

Table 3.1 Criteria of basic emotion theory for the empirical study of emotion (cf. Keltner, Sauter, Tracy, & Cowen, 2019)

<i>How emotions differ from other phenomena</i>	<i>What differentiates emotions from one another</i>
Brief duration	Distinct primate homologies
Unbidden	Distinct physiological correlates
Coherence among response systems	Distinct universal antecedents
Quick onset	Distinct reported experiences
Presence in other primates	Distinct universal signals
Physiological correlates	Distinct behavioural tendencies
Particular antecedent	Distinct appraisal tendency
Reported experience	
Particular signal	
Behavioural tendency	
Cognitive focus	

(Hanin, 2000) recommends a highly individualised approach when dealing with emotions in sports. Although I acknowledge that the classification of emotional states is still a hotly debated topic, I consider the criteria of basic emotion theory in Table 3.1 a promising starting point in giving structure to further empirical research on this topic. Despite the controversy on the classification of emotions, most scientists (reviewed in Ekman, 2016) agree on the two most important functions of emotions: to prepare the organism to respond adaptively to recurring stimuli, and to communicate important social information (Darwin, 1872; see Shariff & Tracy, 2011). In this chapter, I will focus on the communicative function of emotions as this seems to be a fairly neglected aspect of emotion research in sports that offers a lot of potential for gaining important insights on the relationship between emotions and performance as “what we see outside tells us a lot about what’s happening inside an individual’s head and heart” (Beckoff, 2007, p. 1).

The Expressions of Emotions

Darwin (1872) is often credited for launching the scientific study of emotion expressions with his seminal book *On the Expressions of the Emotions in Man and Animal*. With his astounding powers of observation, Darwin carefully documented that certain internal states, such as emotions, are universally expressed, which he considered an example of an evolved class of behaviours that serve adaptive functions for individuals and species. His assumption was that these expressions are vestiges of serviceable associated habits (i.e. behaviours that used to have specific adaptive functions). For example, baring the teeth is a prerequisite for an assault amongst species that attack with their teeth; or wrinkling the nose reduces the inhalation of harmful odours. According to Darwin, the formerly functional expressions continued to persist in humans, although they no longer serve their original purpose. Or stated differently, the expressions emancipated themselves from their original biological functions (Tinbergen, 1952). In the course of evolutionary history, expressive behaviours acquired communicative value as they provided others with external evidence of an individual’s internal state. “The facial expressions we observe today are a rich shorthand for communicating the possibility of more full-bodied actions—attack, flight, embrace” (Keltner, 2009, p. 22).

In a general sense, emotion expressions are considered an instance of non-verbal behaviour (NVB). The importance of NVB in humans is often highlighted with Watzlawick, Beavin, and Jackson’s (1967, p. 51) aphorism “one cannot not communicate” as people are constantly sending out nonverbal signals that convey some kind of message to observers. Studies suggest that the vast majority of information exchanged between individuals is conveyed nonverbally, although estimates of the exact percentages are believed to range between 65–95% (see Matsumoto, Frank, & Hwang, 2013 for a review). It is important to note that NVB can convey many other kinds of information besides emotions, such as information relevant to opinions, values, personality

dispositions, psychopathologies, physical states such as fatigue, and cognitive states such as comprehension or interest (Fridlund, 1994). Hence, the theoretical notion that NVB – particularly faces, but also vocal, postural expressions, touch, proxemics, and gaze – directly express emotions has come under increasing scrutiny (Crivelli & Fridlund 2018). According to this view, NVB is assumed to communicate how people are likely to act, rather than how they are currently feeling. I will not go into this debate but rather circumvent it by settling on the diplomatic consensus that NVB can convey accurate information about internal states of a person: “Indeed, many emotions are precisely forms of social motives, and many social motives are emotional” (Parkinson 2005, p. 301). Hence, NVB informs others about a person’s current state (including emotional information) and how he or she is likely to behave in the near future.

Although a large body of theoretical and empirical work suggests that NVB often autonomously conveys accurate information about internal states of people, it is important to note that people (in contrast to most other species) can also “fake” or deliberately modify certain nonverbal expressions (DePaulo, 1992). Hence, NVB is conceptualised to systematically vary along a continuum of controllability (Ekman & Friesen, 1969) with some nonverbal displays being under conscious, deliberate control, and others under unconscious, autonomous control. Goffman (1959) was one of the first to explicitly state that people can purposely control their NVB to convey particular impressions. In this respect, one of the most important questions on NVB in sports is the question of when NVB can and cannot be deliberately produced and when it might leak information that athletes or officials would prefer to deliberately hide. Typically, people assume that desired nonverbal expressions can be produced whenever and without any problems (DePaulo, 1992). However, this is often more difficult than one assumes given the evolutionary origin of emotion expressions. For example, if professional sport referees want to convey an impression of competence when communicating a decision in a high-pressure situation, they might instruct themselves to stand erect, hold their head high, directly look at the athlete, and speak calmly in an unwavering voice. Unfortunately for referees, research has shown this might not be as easily done during competition as desired (Furley & Schweizer, 2016b), as their NVB has been shown to “leak” signs of insecurity when communicating difficult decisions and thereby potentially influence subsequent social interactions.

In general, the display and observation of certain NVBs in social interactions have been shown to have a considerable impact on a variety of subsequent interpersonal outcomes, including cognition, emotion, behaviour, and performance (see Matsumoto et al., 2013 for a review). To better understand how (emotional) NVB may exert these interpersonal effects, the *emotions as social information model* (EASI-model; van Kleef, 2009) was proposed, assuming that NVB affects observers via two routes: inferential processes and/or affective reactions. Inferential processes describe how an

observer of NVB is able to infer certain information about internal states (e.g. feelings, attitudes, relational orientations) of other people and use this information to better evaluate the situation. On the other hand, the observed expressions can directly elicit affective reactions within the observer via the process of emotional contagion (Hatfield, Cacioppo, & Rapson, 1993, see also Chapter 5).

Selected Research on Emotion Expressions in Sports

Several studies suggest that NVB differs following successful or unsuccessful sports performances (Furley & Schweizer, 2019 for a review). These non-verbal emotional reactions towards success or failure tend to “leak,” meaning that athletes cannot entirely control their nonverbal reactions, even if they try (Furley & Schweizer, 2016b). These findings are in line with evolutionary theorising on emotional nonverbal expressions (Furley, 2019) as they suggest that (1) as a result of winning or losing in antagonistic encounters, humans change their NVB in order to communicate dominance and pride (i.e. high status) or submission and shame (i.e. low status), and that (2) other humans are able to detect and interpret these changes (Furley & Schweizer, 2014, 2016a). These findings have been interpreted as being the result of adaptations that evolved to solve specific adaptive problems in our ancestral past: sending signals when losing an antagonistic encounter increased the chances of avoiding further potentially life-threatening attacks. Likewise, sending victorious signals helped to save valuable resources by communicating high status and superiority over the opponent, and further signal supremacy to potential mates. This assumption is supported by results showing that leading athletes are perceived to be higher on dimensions related to social status (i.e. dominance, pride, confidence) than trailing athletes (Furley & Schweizer, 2016a).

Few studies have investigated the link between NVB and sports performance. For example, in penalty shoot-outs, certain celebratory emotion expressions by successful penalty-takers tentatively predicted the success of the next penalty taken by a member of the opposing team and the likelihood of the celebrating penalty-taker being in the winning team (Moll, Jordet, & Pepping, 2010). Further research suggests that higher pressure during a penalty shootout is related to avoidance NVB, such as looking away from the keeper and speeding through the preparation before the kick (Furley, Dicks, Stendtkke, & Memmert, 2012) which, in turn, is linked to worse performance (Jordet & Hartmann, 2008). A further series of studies on emotional expressions in sports suggest that emotion expressions do not necessarily directly influence sports performance, but emotional displays affect thoughts, motivation, confidence, and emotions of opponents and teammates and in this way indirectly influence performance (Furley & Schweizer, 2019).

Applied Recommendations

Since emotion expressions can influence opponents and teammates, I intend to conclude this chapter with some general practical guidelines for coaches and athletes. While much has been written on emotion regulation (Gross, 2014), more research is needed on the regulation of emotion expressions in sports. Trying to regulate emotions will likely also impact on emotional expressions as emotional expressions are usually closely linked with the emotion and the feeling. This implies that it is difficult to change only one emotional facet without the other (i.e. physiological reaction, emotional expression, and feeling). For example, a very nervous athlete is unlikely to benefit from simply working on trying to hide nonverbal signs of insecurity. Hence, it is not sufficient to simply look up NVBs that have been linked to desirable impressions and try to copy these in certain sport contexts. Nevertheless, it seems worth mentioning that specific feedback loops (e.g. *facial feedback*, *power posing*) have been suggested (albeit this is a controversial topic) to enable humans to use NVB to induce certain emotions and/or feelings (see Furley & Schweizer, 2019).

As an initial step in regulating emotion expressions in sports it seems important to educate people (1) on the assumed evolutionary origins of NVB; (2) on the fact that NVB has some innate components associated with hard-wired emotional modules; (3) on the assumption that it varies along a continuum of controllability; and, finally, (4) on the fact that there are many constraints on the deliberate use of NVB for self-presentational purposes. Given that people are often not aware of their NVB or are mistaken about how they believe other people perceive their NVB, an advisable second step would be to increase awareness of the displayed NVB in certain situations by informing athletes about how their body language is perceived and what reactions it might elicit on observing opponents and teammates. This could, for example, be done with video recordings during competitions in combination with team discussions. In this way, it might be possible to identify instances when NVB diverges from intended self-presentation, and subsequently to attempt to modify NVB in a beneficial way. To this end, it should be possible to shape deliberate practice activities to improve NVB. However, as with every other skill, the deliberate practice activities on NVB have to be thoughtfully developed and practised over an extended period of time for them to be successful.

Conclusion

To fully understand human behaviour, it seems inevitable to approach it from an evolutionary perspective. I argue that the field of emotion research can benefit from adopting an evolutionary perspective. I exemplify this by focusing on the expression of emotions as one aspect of emotions that offers potential for gaining important insights on emotions in sports. Taken together,

a growing body of research shows that NVB plays an important role in sports. In this respect, researchers on emotion expressions in sports would benefit from investigating potential links between NVB and psychological momentum in sports competitions. It seems like a viable hypothesis to assume that emotion expressions on the playing field could prove to be an important mosaic piece in explaining the upward or downward spirals. For example, when it is going poorly in a contest, this shows in an athlete's NVB, causing opponents to perform better and the athlete to perform worse.

Given the overwhelming documentation of sports in combination with its intense emotional nature, and the fact that “what we see outside tells us a lot about what’s happening inside” (Beckoff, 2007), makes sports highly suitable for the study of human emotions. Hence, it is not surprising that emotional expressions recorded during sports competitions have led to important new basic insights in emotion theory (e.g., Aviezer, Trope, & Todorov, 2012). In this respect, it is my hope that evolutionary research on emotion in sports will help further understanding of Tinbergen’s seminal four questions that “are the most important tools in an evolutionary scientist’s repertoire and have to be answered in interdisciplinary research endeavours to fully understand any product of evolution” (Wilson, 2019, p. 37): first, what is/are the function(s) of emotions? Second, what is the history of emotions as they evolved over multiple generations? Third, what are the physical mechanisms of emotions? Fourth, how do emotions develop during the lifetime of the organism?

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